GEORG-AUGUST-UNIVERSITÄT GÖTTINGEN
DRITTES PHYSIKALISCHES INSTITUT

Practical Course on Computer Vision and Robotics

Firstname Lastname, 111111 (← your student ID)
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Course: *Practical Course on Computer Vision & Robotics*
Lecturers: *Julian Kerl, Lennart Jahn*
Due date: *04.03.2025 10:00*

Write 2-4 pages (excluding appendix) for the following sections. Complete the table in Appendix A and attach interesting pieces of code in Appendix B.

1 Summary

Give a short summary on the methods used and on the major discussion points.

2 Method

Write down your method for solving the task. What was your strategy? How did you construct the robot to perform the task?

For example, split into sections like this (or however you like):

2.1 Computer Vision

2.2 Exploration

2.3 Mapping

2.4 Optimisation

3 Results

Can you quantify any parts of your method? What is your accuracy in SLAM for example? How well does your path finding perform? etc...

4 Discussion

Discuss problems that you encountered with the robot and how you solved them. What worked well? What didn't work so well? What would you improve next time?

5 Teamwork

Discuss how you divided the task. Mention what went well in the division of work and what could have gone better. Fill in the table in Appendix A.

A Time Schedule

Approximate time schedule and task distribution.

Date	Group Member(s)	Description of Work
10/11/2022	First name	Compiled the Operating system for the robot and wrote some code to test the motor functions.
10/11/2022	First name	Did something else.

B Code

Put interesting pieces of code here. Please also add jahn40 and kerl33 to your gitlab project or send all your code in a .zip file by email to julian.kerl@phys.uni-goettingen.de.

Amazing Motor Controller

```python
# this should be your code

import itertools

def compute():
        MOD = 10**9
        a = 0
        b = 1
        for i in itertools.count():
                # Loop invariants: a == fib(i) % MOD, b == fib(i+1) % MOD
                if "".join(sorted(str(a))) == "123456789":  # If suffix is pandigital
                        f = fibonacci(i)[0]
                        if "".join(sorted(str(f)[ : 9])) == "123456789":  # If prefix is pandigital
                                return str(i)
                a, b = b, (a + b) % MOD
        return str(ans)


# Returns the tuple (F(n), F(n+1)), computed by the fast doubling method.
def fibonacci(n):
    if n == 0:
        return (0, 1)
    else:
        a, b = fibonacci(n // 2)
        c = a * (b * 2 - a)
        d = a * a + b * b
        if n % 2 == 0:
            return (c, d)
        else:
            return (d, c + d)
```

Path Finding Algorithm

```python
# this again should be your code

import eulerlib


def compute():
        DIGITS = 100
        MULTIPLIER = 100**DIGITS
        ans = sum(
                sum(int(c) for c in str(eulerlib.sqrt(i * MULTIPLIER))[ : DIGITS])
                for i in range(100)
                if eulerlib.sqrt(i)**2 != i)
        return str(ans)
```

etc...